Inference for Categorical Time Series with Missing Observations

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Outline

Motivation

- Many real-world time series are **categorical**:
  - Credit ratings (AAA, AA, ...)
  - Weather states (sunny, rainy, ...)
  - Survey responses (agree, neutral, disagree)
- Classical time series methods assume **numeric structure**
- $\Rightarrow$ Need new tools for:
  - Computing proper tests for marginal features
  - Measuring temporal dependence without bias

Categorical Time Series

Definition

A categorical time series (X_t) takes values in a finite set

$$S = \{s_0, \ldots, s_m\},$$

e.g. $X_t \in S = \{$  ,  ,  $\}$

Serial Dependence

In many applications, observations are not independent over time:

- The distribution of X_t may depend on past values $X_{t-1}, X_{t-2}, \ldots$

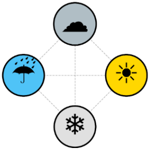
Key implication:

- Available statistical tools depend on both the structure of S and the dependence over time

Ordinal vs. Nominal Data

Nominal data (e.g. eye color, gender):

- No natural ordering of categories



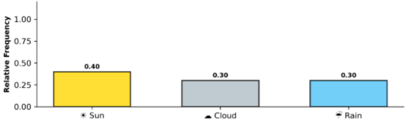
Ordinal data (e.g. education level, satisfaction ratings):

- Elememts of S have a natural ordering
- However, intervals between categories are not meaningful

Marginal Distributions

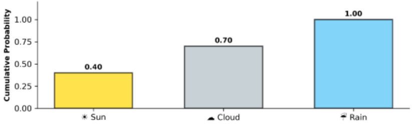
PMF(Probability Mass Function):

$$p_i = P(X_t = s_i)$$
$$\Rightarrow \boldsymbol{p} = (p_0, p_1, \ldots, p_m)^\top$$



CDF(Cumulative Distribution Function):

$$f_i = P(X_t \leq s_i)$$
$$\Rightarrow \boldsymbol{f} = (f_0, f_1, \ldots, f_m)^\top$$



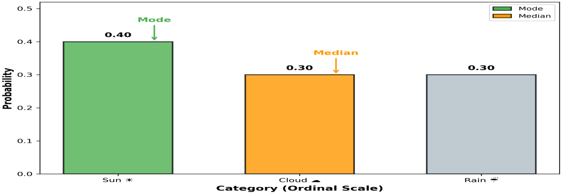
Location Measures

Mode:

$$\mathrm{Mod}(X_t) = \arg\max_{s_i} P(X_t = s_i)$$

Median:

$$\mathrm{Med}(X_t) = \min\{s_i : F(s_i) \geq 1/2\}$$

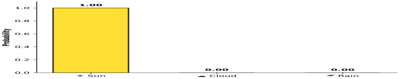


Interpretation:

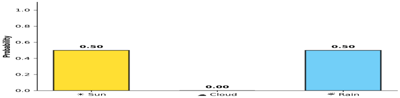
- Mode = most frequent category
- Median = central position in ordering

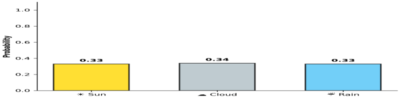
Dispersion

Minimal Dispersion:



Maximal Dispersion:





Ordinal

$$\mathrm{IOV}(\boldsymbol{f}) = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i)$$

Nominal

$$\mathrm{IQV}(\boldsymbol{p}) = \frac{m+1}{m} \left(1 - \sum_{i=0}^{m} p_i^2\right)$$

From here on: We focus on **ordinal data**, unless stated otherwise.

Stationarity

Why do we need stationarity?

- Stable probabilistic structure over time
- Enables estimation and asymptotics

Strict Stationarity

$$F_{X_t,\ldots,X_{t+j}} = F_{X_{t+h},\ldots,X_{t+h+j}}$$

for all t, j, h.

- Distribution invariant under time shifts

Temporal Dependence

Goal: capture dependence across time

Problem:

- Classical ACF not applicable

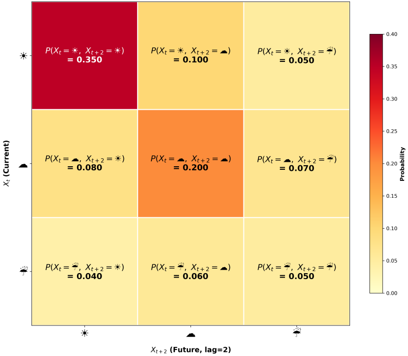
Solution:

Bivariate Probabilities

$$p_{ij}(h) = P(X_t = s_i, X_{t+h} = s_j)$$
$$f_{ij}(h) = P(X_t \leq s_i, X_{t+h} \leq s_j)$$

Bei Unabhängigkeit:

$$p_{ij}(h) = p_i p_j \quad \text{und} \quad f_{ij}(h) = f_i f_j$$



Cohen's κ

Idea: Standardized measure of deviation from independence

$$\kappa(h)_{\mathrm{ord}} = \frac{\sum_{j=0}^{m}(f_{jj}(h) - f_j^2)}{1 - \sum_{i=0}^{m} f_i(1 - f_i)}$$

- $\kappa = 1$: perfect persistence
- $\kappa = 0$: independence
- $\kappa < 0$: tendency to switch states

Interpretation:

- Probability of staying in same category
- Relative to independence benchmark

From Theory to Estimation

All quantities defined so far are population quantities.

Central question

How do we estimate $\boldsymbol{f}$, IOV, and $\kappa_{\mathrm{ord}}(h)$ from data — and what are the statistical properties of these estimators?

Two steps:

1. Estimate marginal CDF $\boldsymbol{f}(/\boldsymbol{p})$ (and later the bivariate probabilities) via relative frequencies
2. Derive (asymptotic) distributions of plug-in estimators using a **CLT** and the **delta method**

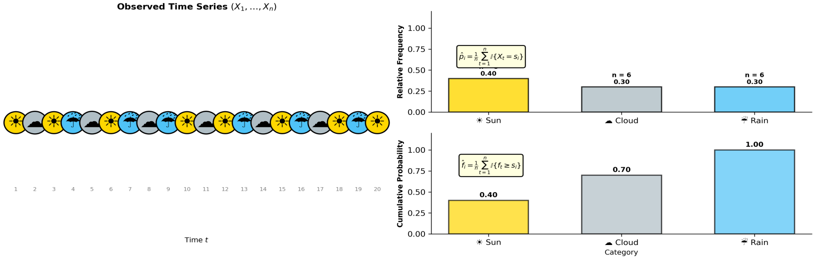
Estimation of Marginal Probabilities

Idea: use of indicator functions

$$Y_{t,i} = \mathbb{I}\{X_t = s_i\} \quad \text{and} \quad Z_{t,i} = \mathbb{I}\{X_t \leq s_i\}$$

Then estimators are given by:

$$\hat{p}_i = \frac{1}{n}\sum_{t=1}^{n} Y_{t,i} \quad \text{and} \quad \hat{f}_i = \frac{1}{n}\sum_{t=1}^{n} Z_{t,i}$$



Deriving Asymptotic Distributions

Goal

We aim to derive asymptotic distributions for estimators based on categorical time series for inference.

Key tool

The **Central Limit Theorem (CLT)** provides asymptotic normality:

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, \sigma^2)$$

Problem

Classical CLTs require **independent observations**, but time series data typically exhibit **serial dependence**:

- Current observations depend on past values
- This dependence persists across time lags

Mixing Conditions

Idea

We control serial dependence via **mixing conditions**, which quantify how fast dependence between past and future decays.

α-mixing (strong mixing)

Let $\mathcal{F}^t_{-\infty} = \sigma(X_s : s \leq t)$ and $\mathcal{F}^{\infty}_{t+h} = \sigma(X_s : s \geq t + h)$.
The α-mixing coefficients are defined as:

$$\alpha(h) = \sup_{t\in\mathbb{Z}} \sup_{A\in\mathcal{F}^t_{-\infty},\, B\in\mathcal{F}^{\infty}_{t+h}} \left| P(A \cap B) - P(A)P(B) \right|$$

- $\alpha(h) \overset{h\to\infty}{\to} 0$: dependence between past and future vanishes
- Exponential decay $\Rightarrow$ CLT for dependent data

CLT under Serial Dependence

Result

$$\sqrt{n}(\hat{\boldsymbol{f}} - \boldsymbol{f}) \Rightarrow \mathcal{N}(0, \boldsymbol{\Sigma})$$

where the elements of $\boldsymbol{\Sigma}$ are given by:

$$\sigma_{ij} = \underbrace{\left(f_{\min\{i,j\}} - f_i f_j\right)}_{\text{lag}(0)} + \sum_{h=1}^{\infty} \underbrace{\left(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\right)}_{\text{lag}(h)}.$$

Note that $\text{lag}(h) \to 0$ as $h \to \infty$.

CLTs for Marginal Estimators

Idea

IOV is a function of the marginal probabilities
$\Rightarrow$ apply the **delta method** to obtain asymptotic distributions

Results

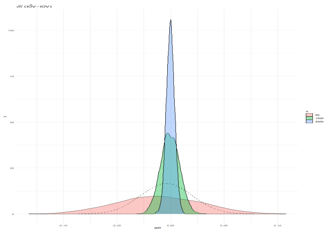
- Asymptotic variance:

$$\mathrm{Var}(\sqrt{n} \cdot \widehat{\mathrm{IOV}}) \stackrel{\mathrm{asy}}{=} \frac{16}{m^2} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j)\sigma_{ij}$$

- Bias to order n^{-1}:

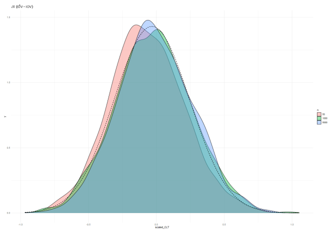
$$\mathrm{Bias}(n \cdot \widehat{\mathrm{IOV}}) = -\frac{4}{m} \sum_{i=0}^{m-1} \sigma_{ii}$$

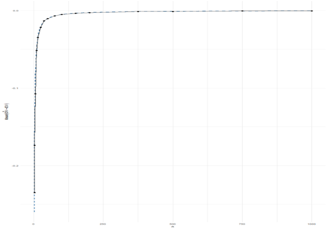
Scaling of Variance and Bias



Variance

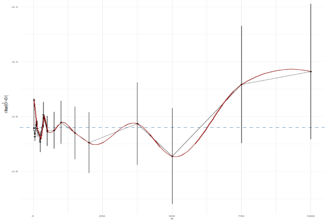
$\xrightarrow{\cdot\sqrt{n}}$





Bias

$\xrightarrow{\cdot n}$



Why Does Missingness Matter?

Missing observations are ubiquitous in practice:

- Panel surveys: participants drop out over time
- Sensor data: transmission errors, device downtime

Problem

- Simply ignoring missing observations leads to biased estimates of standard errors and plug-in estimators

Goal:

- Correct for missingness — under suitable assumptions
- Derive valid asymptotic theory for the corrected estimators

Amplitude Modulation

Idea

Missingness can be modeled as a binary process (O_t), that *modulates* the *amplitude* of the actual process:

$$O_t = 1 \text{ if } X_t \text{ is observed and } O_t = 0 \text{ if } X_t \text{ is missing}$$

The observed process is then given by $(O_t X_t)$.

Assumptions

- (O_t) is stationary, with $P(O_t = 1) = \pi$ and joint moments $E[O_t O_{t+h}] = \pi(h)$ (if O_t is i.i.d., then $\pi(h) = \pi^2$)
- (O_t) is α-mixing with exponential decay

Dependence structure of (O_t) and (X_t)

- **Indepedence:** ($\sim$MCAR)

$$P(O_t = 1 | X_t, X_{t-1}, \ldots) \;= P(O_t = 1)$$

- **Dependendence on past states:** ($\sim$MCAR)

$$P(O_t = 1 | X_t, X_{t-1}, \ldots) \;= P(O_t = 1 | X_{t-1}, \ldots)$$

- **Dependence on current state:** ($\sim$MCAR)

$$P(O_t = 1 | X_t, X_{t-1}, \ldots) \;\neq P(O_t = 1 | X_{t-1}, \ldots)$$

From here on, unless stated otherwise, we assume Independence.

Naive vs. Corrected Estimators

Naive estimator

$$\hat{\mathfrak{f}} = \frac{1}{n} \sum_{t=1}^{n} O_t \boldsymbol{Z}_t$$

$\Rightarrow$ biased and inconsistent: $E[\hat{\mathfrak{f}}_i] = \pi f_i \neq f_i$

Corrected estimator

$$\hat{\boldsymbol{f}}^* = \frac{\frac{1}{n} \sum_{t=1}^{n} O_t \boldsymbol{Z}_t}{\frac{1}{n} \sum_{t=1}^{n} O_t}$$

$\Rightarrow$ divide by number of observed time points, not by n

Asymptotic Properties of the Corrected Estimator

Result

$$\sqrt{n}(\hat{\boldsymbol{f}}^* - \boldsymbol{f}) \Rightarrow \mathcal{N}(0, \boldsymbol{\Sigma}^*)$$

where the elements of $\boldsymbol{\Sigma}^*$ are given by:

$$\sigma_{ij}^* = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2} \sum_{h=1}^{\infty} \pi(h)\big(f_{ij}(h) + f_{ji}(h) - 2 f_i f_j\big)$$

Remarks:

- Inflation factor $1/\pi$ at lag 0: fewer effective observations
- Serial dependence in (O_t) enters via $\pi(h) = E[O_t O_{t+h}]$
- Reduces to $\boldsymbol{\Sigma}$ when $\pi = 1$ (no missing data)

Marginal Statistics under Missingness

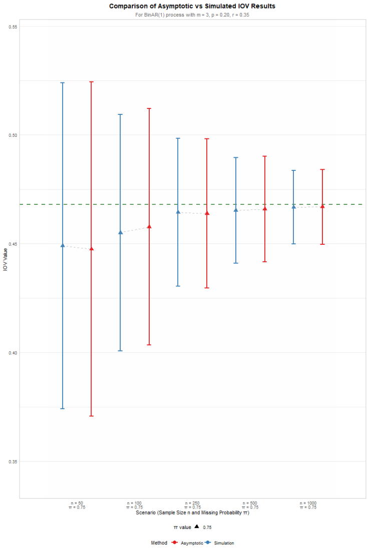
Strategy

All results from the complete-data case carry over by replacing $\boldsymbol{\Sigma}$ with the missingness-adjusted covariance matrix $\boldsymbol{\Sigma}^*$.

IOV: Same delta-method expansion as before, with $\boldsymbol{\Sigma}$ replaced by $\boldsymbol{\Sigma}^*$.

Other statistics (e.g. skewness): The same delta-method arguments apply.

IOV under Missingness



- Bias decreases linear in n
- Variance estimates approach the asymptotic variance as *n* increases
- Asymptotic bias formula captures the finite-sample bias well, even for moderate *n*

Estimating Bivariate Probabilities

Need to estimate $f_{ij}(h) = P(X_t \le s_i, X_{t+h} \le s_j)$

Corrected estimator:

$$\hat{f}^*_{ij}(h) = \frac{\frac{1}{n-h}\sum_{t=h+1}^{n} O_t\, O_{t-h}\, Z_{t,i}\, Z_{t-h,j}}{\frac{1}{n+h}\sum_{t=h+1}^{n} O_t\, O_{t-h}}$$

$\Rightarrow$ divide by the proportion of **jointly observed** ($\frac{1}{n+h}\sum_{t=h+1}^{n} O_t\, O_{t-h}$) pairs at lag h

Asymptotic Distribution of $\hat{\kappa}_{\mathrm{ord}}(h)$

CLT + Delta method: The corrected estimators of $\boldsymbol{f}$ and $\boldsymbol{f}(h)$ are jointly asymptotically normal, hence (continuous mapping theorem)

$$\sqrt{n}\left(\hat{\kappa}_{\mathrm{ord}}(h) - \kappa_{\mathrm{ord}}(h)\right) \xrightarrow{d} \mathcal{N}(0, \sigma_\kappa^2)$$

Challenge: Higher-order moments of (X_t) are in σ_κ^2.

$\Rightarrow$ We therefore consider the **null hypothesis of i.i.d. observations**, where $f_{ij}(h) = f_i f_j$, and test on deviation from that.

Asymptotic Null Distribution of $\hat{\kappa}_{\mathrm{ord}}(h)$

Theorem

Under H_0 and the previous assumptions on (O_t) and X_t, we can compute σ_κ^2 in closed form:

$$\sigma_\kappa^2 = \frac{\frac{1}{\pi(h)} \sum_{i,j=0}^{m-1} (f_{\min\{i,j\}} - f_i f_j)\left(f_{\min\{i,j\}} + \left(1 + \frac{2\pi(h,2h)}{\pi(h)} - \frac{4\pi(h)}{\pi}\right) f_i f_j\right)}{\left(\sum_{k=0}^{m-1} f_k(1-f_k)\right)^2}$$

with $\pi(h, 2h) = E[O_t O_{t+h} O_{t-h}]$.

Bias Correction for $\hat{\kappa}_{\mathrm{ord}}(h)$

Second-order Taylor expansion yields the finite-sample bias under H_0:

$$\mathrm{Bias}(n\hat{\kappa}_{\mathrm{ord}}(h)) \approx -\frac{1}{\pi}$$

Bias-corrected estimator:

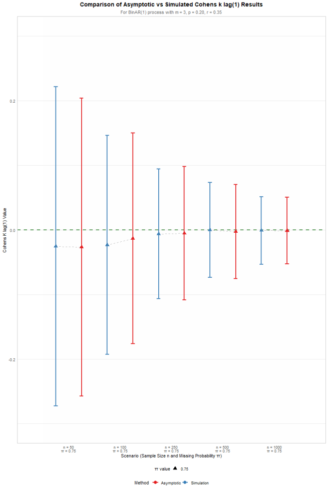
$$\hat{\kappa}^{\mathrm{bc}}_{\mathrm{ord}}(h) := \hat{\kappa}_{\mathrm{ord}}(h) + \frac{1}{n\pi}$$

Remarks:

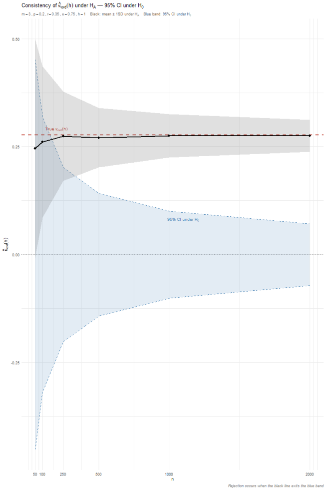
- Bias increases as $\pi \downarrow$ (more missingness)
- Under complete data ($\pi = 1$): reduces to the known $-1/n$ bias

Asymptotic Distribution of $\hat{\kappa}_{\text{ord}}(h)$

Under H_0



Under H_A(95% confidence intervals)



Simulation Goals and Setup

Three goals:

1. Assess finite-sample accuracy of asymptotic distributional properties
2. Examine convergence under serially dependent missingness
3. Study robustness to violations of the MCAR assumption

Data Generating Process: BinAR(1)

Goal: Simulate ordinal time series with controllable dependence.

Idea: A discrete autoregressive process on $\{0, \ldots, m\}$ with binomial transitions.

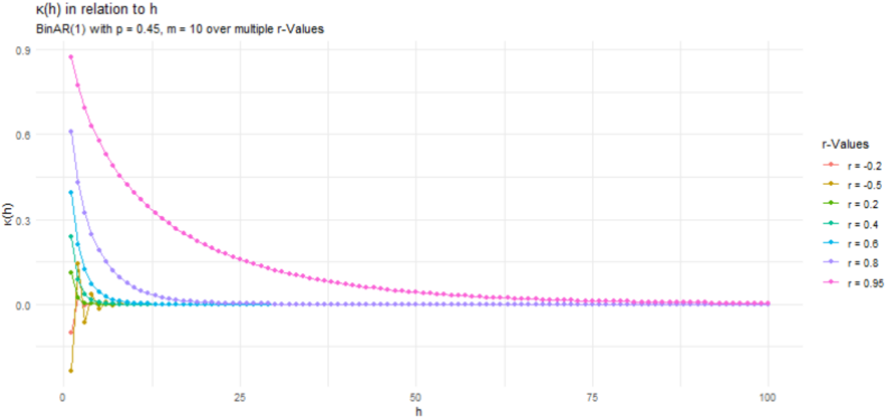
Definition (intuition):

$$X_t = B(X_{t-1}, \alpha) + B(m - X_{t-1}, \beta)$$

- $X_t \in \{0, \ldots, m\}$ (ordinal / categorical structure)
- p controls the marginal distribution: $X_t \sim \mathrm{Bin}(m, p)$
- r controls serial dependence
  - $r = 0$: i.i.d. case
  - $r > 0$: positive dependence
  - $r < 0$: negative dependence

$\Rightarrow$ Flexible benchmark process with closed-form marginals and dependence structure.

BinAR(1) with varying degree of serial dependence



Simulation Scenarios

Two scenarios chosen to differ qualitatively:

Scenario A

$m = 3,\ p = 0.20,\ r = 0.35$

- Few categories
- Skewed marginal distribution
- Moderate serial dependence

Scenario B

$m = 10,\ p = 0.45,\ r = 0.50$

- Many categories
- Near-symmetric distribution
- Strong serial dependence

Design:

- Observation rates: $\pi \in \{1, 0.75\}$
- Sample sizes: $n \in \{50, 100, 250, 500, 1000\}$
- $B = 1{,}000$ Monte Carlo replications

Missingness Mechanisms in the Simulation

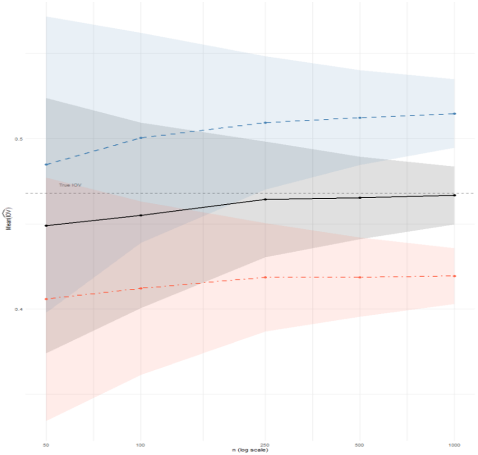
Four observation processes (O_t) are studied:

1. $\boldsymbol{O_t} \perp \boldsymbol{X_t}$ **and no serial dependence**: $O_t \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\pi)$
   Baseline; satisfies all theoretical assumptions.

2. $\boldsymbol{O_t} \perp \boldsymbol{X_t}$ **and serial dependence**: (O_t) follows BinAR(1) with $m = 1$
   Serial dependence in missingness.

3. **Dependence between O_t and X_t**: Misssingness is state-dependent
   Higher/ Lower Categories are more likely to be observed.

MNAR vs. MCAR: IOV Bias



MNAR — increasing, MNAR — decreasing

and MCAR

- MNAR is biased and not consistent

Outlook

Remaining work

- Nominal data — transfer of results from ordinal to nominal case
- Simulation study — finite-sample validation of asymptotic results of $\kappa(h)$
- Analysis of impact of violations of the MCAR assumption

Open research questions

- Asymptotic theory given dependence of (O_t) on past states of (X_t) (MAR)
- Extensions — multivariate categorical processes
- Applications — credit ratings, longitudinal survey data